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New unified physical model for surface integrity prediction considering the state of stress and the microstructure effects

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Abstract

The presented work targets the development of a numerical orthogonal cutting model for surface integrity prediction. It incorporates a new unified physical model of OFHC copper. Due to their importance in the material behavior, the effects of the state of stress on the flow stress evolution up to fracture and the microstructure of the work material are taken into account. This formulation brightens also the microstructure effects on the flow stress. Orthogonal cutting model is validated using experimental designed cutting tests with different cutting conditions, including different tool geometries. The obtained predictions using this unified physical model are then compared to the classical Johnson-Cook constitutive model.

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Keywords: Predictive model; surface integrity; cutting; finite element method (FEM); OFHC copper.
